233/1

CHEMISTRY

PAPER 1.

MARKKING SCHEME

a) It varies according to traffic flow.✓ ½ when the flow is low, vehicle move fast, combustion of petrol is almost complete and hence carbon(II)oxide proportional is 60pp.m and when flow is high, vehicle move slowly, at time stop ✓ ½ and start engine, high concentration of carbon(II) oxide which can be 180p.pm

b) Incomplete combustion ✓ ½ of petrol due to limited ✓ ½ supply of oxygen (air) in engine

c) Takes site of oxygen in red blood cells to form carboxyhaemoglobin ✓ ½ which does not decompose. Therefore blood has less oxygen leading✓ ½ to reduced respiration (suffocation)

2. a) Saponification reaction ✓01

b) Fatty acids ✓01

c) -Add brine ✓ ½ (Conc. NaCl) to acquires mixture, M precipitates out leaving L in solution. Decant ✓ ½ off L to separate from M

3. a) Nitrogen gas ✓ 01

b) Remove delivery tube from under water ✓01 while still heating to prevent suck back. If heating is stopped while still under water, water will rush back into the test tube since atmosphere pressure on surface of water is greater than pressure in the tube. ✓01

4. i) J ✓01
ii) K ✓01
iii) M ✓01

5. a) i) Oxidation $-Pb_{(s)} \longrightarrow Pb^{2+}_{(aq)} + 2e^-$ ✓01
ii) Reduction $Cu^{2+}_{(aq)} + 2e^- \longrightarrow Cu_{(s)}$ ✓01

reject
- Joining of letters
- Proportionality of celters.
- Oxidation numbers instead of charge.

b) e.m.f = E^θ reduced - E^θ oxidized.
= + 0.34 – (-0.13) ✓ ½
= +0.47v ✓ ½

6. a) frome equation 1mole CH_4 gives 206KJ

? ______ 556KJ

$\frac{556x1}{206}$ = 2.699moles ✓01

1 mole has a mass of 16g

2.699 mol………….. ?

$\frac{2.699x16}{1}$ = 43.43.18g ✓01

b) Reduced yield ✓01 of carbon(II)oxide. Production of carbon (II) oxide is favored by high pressure therefore by le-chateliers principle increase in pressure shifts equilibrium to the left. ✓ ½

7. a) Mole of $BaSO_4$ $= \frac{Mass}{R.F.M}$

$= \frac{4.66}{152}$

=0.0306moles ✓ ½

$Ba^{2+}{}_{(aq)} + SO^{2-}{}_{(aq)} \rightarrow BaSO_{4(s)}$ ✓ ½

Moles of SO_4^{2-} = moles of $BaSO_4$ = 0.0306 moles ✓ ½

b) $M_2SO_{4(aq)} \rightarrow 2M^{+}{}_{(aq)} + SO_4^{2-}$ = 2:1 ✓ ½

Therefore moles of M^+ = 2x0.0306 = 0.0612 moles,

Mass of SO_4^{2-} = in M_2SO_4 = 96 x 0.0306

= 2.94 g ✓ ½

∴ Mass of M in M_2SO_4= 5.34-2.94= 2.4g ✓ ½

Mass $= \frac{Mass}{R.F.M}$ ∴ R.A.M of M= $\frac{2.4}{0.0612}$

=39.2 ✓ ½

8. -Add the mixture to ✓01 either and filter to remove aluminum sulphate and sugar residue.
-Add residue to alcohol ✓ 01and filter to remove aluminum sulphate. Evaporate ✓01 the alcohol in filtrate to obtain sugar crystals.

9. Total energy change for $Fe_{(g)}$ = 15.4 + 354 = 369.4

369.KJ ✓ ½

To change 1 mole from $Fe_{(s)}$ to $Fe_{(g)}$ require 369.4KJ

To change 0.2 mol from $Fe_{(s)}$ + $Fe_{(g)}$ require?

$\frac{369.4x0.2}{1}$ ✓ ½ = 73.88KJ✓01

Or

To change 56g from solid Fe to gas → require 369.4KJ ✓ ½

To change 11.2g from solid to gas require?

$\frac{369.4x11.2}{56}$ ✓ ½ = 73.88 KJ ✓01

10. i) X -Oxygen gas ✓ 01

P -Water. ✓01

11. a) For reaction to occur, F is reduced and C is oxidissed.

$\therefore$ e.m.f = E^θ reduced - E^θ oxidized.

= + 0.8 –(-0.4)

= +1.2V ✓ 01

Reaction occurs since E^θ is positive. ✓01

b) $B^+_{(aq)} + e^- \rightarrow B_{(s)}$ E^θ = +1.68V ✓01 and

$E^{2+}_{(aq)} + 2e^- \rightarrow E_{(s)}$ E^θ = -2.38v

N/B must indicate E^θ value.

12. Reaction will be faster.✓01 powdered magnesium offers larger surface area ✓01 leading to higher contact of reacting particles or more collisions 01 of reacting particles.

13. A- Nail is covered with brown substance ✓ ½ or Rust because copper is less reactive than ✓ ½ iron, so iron rusts because copper cannot go into solution ✓ ½ in preference to iron.

B- No observable change .✓ ½ zinc is more reactive ✓ than iron, it goes in solution leaving iron ✓ ½ inactive. Zinc is a sacrificial metal.

14. Na_2CO_3 does not decompose, $NaHCO_3$ decompose as :

$2NaHCO_{3(s)} \rightarrow Na_2CO_{3(s)} + H_2O_{(l)} + CO_{2(g)}$ ✓ ½

Mass of $H_2O + CO_2$ =(5.04 - 4.11)g

=0.93g ✓ ½

From equation, 2mooles of $NaHCO_3$ give 1 mole(H_2O+CO_2)

Or 168_s of $NaHCO_3$ ____________ 1mole ($H_2O + CO_2$) = 62g

$\frac{168 \times 0.93}{62}$ =2.52g ✓ ½

Mass of anhydrous Na_2CO_3 = (5.04 – 2.52)g

= 2.52g ✓ ½

% of anhydrous Na_2CO_3 = $\frac{\text{Massof}Na_2CO_3}{\text{Massofmixture}} \times 100\%$

= $\frac{2.52}{5.04} \times 100\%$ ✓ ½

= 50 % ✓ ½

15. Misty fumes, ✓ ½ effervescence ✓ ½ and sulphur dissolves ✓ ½ concentrated sulphuric acid oxidizes ✓1 sulphur to sulphur(IV)oxide gas

Equ! -1

16. a) Curve I. ✓ 01 concentration of F increase with time since it is the product. ✓01

b) After time 'T' concentration of bolt E and F is constant because equilibrium has been established ✓01

17. Copper metal – mass increase ✓ ½ because oxygen ✓ ½ combines with copper metal (oxidation)

or

$ZCu_{(s)} + O_{2(g)} \rightarrow 2CUO_{(s)}$ ✓ ½

Copper nitrate – mass decreases. It decomposes ✓ ½ into gases that escape and oxide left.

Or

$2Pb(NO_3)_{2(s)} \rightarrow 4NO_{2(g)} + O_{2(g)} + O_{2(g)} + O_{2(g)} + 2PbO_{(s)}$ ✓ ½

Anhydrous copper (II) sulphate -mass remains constant ✓ ½ it is not decomposed. ½

18. The volume of a fixed mass of a gas is directly proportional to its absolute temperature if pressure is kept constant. ✓ 01

Explanation.

- increase in temperature ✓01 increases kinetic energy of the molecules, increasing number of collusions per unit time or pressure ✓01 increases to keep pressure constant, volume must increase.

19. i) Ammonia forms alkaline solution. ✓01.

ii) Ammonia is very soluble in water therefore it creates a partial vacuum in upper flask and water in lower flak is forced up the tube .✓ 01

iii) Side tube maintains atmosphere pressure along the tube. ✓ 01.

20. a) To occupy space previously occupied by oxygen that was used by burning phosphorous. ✓01

b) Because oxides of phosphorous formed still occupy space enviously occupied by oxygen. (P_2O_5, P_2O_3) ✓01

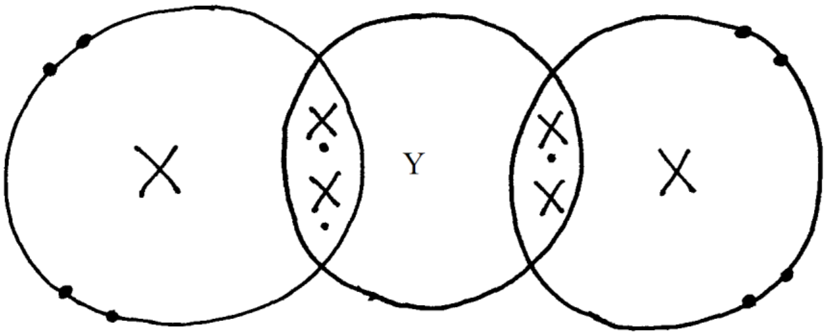
c) Let all the fumes dissolve in water before final reading is taken ✓01

21.

Property	**Rhombic**	**Monoclinic**
Appearance	Bright yellow ✓ ½	Amber coloned ✓ ½
Density g/cm^3	2.08 ✓ ½	1.98 ✓ ½
Melting point	113°C ½	119°C ✓ ½

22. a) 2.8.6 ✓ 01

b) Y has 4 electrons in the outermost energy level, therefore compound X and Y is ✓01



23. number of electrons in the uttermost ✓01 energy level determines group in which it belongs and electronic configuration which gives ✓01 period as it is same as number of energy levels

24. pH will be above seven ✓ ½ 2M sodium hydroxide has more ions than an equal volume of 2M ethanoic acid ✓ ½ it is fully dissociated in solution unlike ethanoic acid which is partially dissociated .All hydrogen ions from acid will be neutralized ✓ ½ by excess hydroxide ions ✓ ½

25. a) Salt X ✓ ½ - its solubility increase with increase in temperature ✓ ½

b) For each salt determine solubility at 90°C and 20°C ✓ 01. For X it reduces, therefore X Crystallizes. For Y it is almost constant, therefore no crystals. ✓01

26. a) Separating funnel ✓ 01

b) B ✓ 01

c) Decanting ✓01 or sacking top of liquid or into another container using teat pipette. ✓01

27. i) L ✓ 01

ii) J ✓01

28.

Compound	**$Ba(OH)_2$**	**nH_2O**
Mass	51.3g	433.25
R.A.M	171	18
Moles	$\frac{51.3}{171} = 0.3$ ✓ ½	$\frac{43.2}{18} = 2.4$ ✓ ½
Mole ratio	$\frac{0.3}{0.3}$ ✓ ½	$\frac{2.4}{0.3}$ ✓ ½

1 8

$Ba(OH)_2 8H_2O 8H_2O$ n=8 ✓ ½

29. Dip the blue litmus paper in each ✓ ½ solution. H_2SO_4 turns it red ✓ ½ while K_2CO_3 and NaCl have no effect ✓ ½ transfer the two unknown solutions to a test tube. Add H_2SO_4 to each solutions ✓ ½ solution with K_2CO_3 give vigorous effervescence ✓ ½ while solution of NaCl give no observable change. ✓ ½